Live demo

The methods on real public data

Status: Draft v0.1 (2026-11-04), for Steve's review. Extension 8 of 10 (see the extensions overview). Code: `analytics/demo/run_demo.py` (scores) and `analytics/demo/fetch_usaspending.py` (pulls awards); results: `docs/demo/data/demo.json`; shown on the site at `/demo/`.

Bottom line

The site's tool pages were wireframes with sample records. The demo replaces "imagine this" with "here it is": the same reference code that the forensic methods handbook describes, run on real public data, with the results on a page. It deliberately starts with a dataset where the honest answer is "nothing to see", because the most important thing to show a leader is that the tests know their own limits.

What runs today, on data committed to the repo:

1. **Digit test on every agency's FY2026 outlays, budget authority and obligations** (USAspending). 301 amounts from 101 agencies.
2. **Concentration** of federal outlays across agencies.
3. **Reconciliation:** agencies whose obligations or outlays exceed budget authority by more than 5%.
4. **The ROI model** on GAO's FY2025 improper payment figures.

What runs where the network allows (the "Demo data" GitHub Actions workflow, or a laptop): contract awards for any agency from the USAspending API, scored with the digit test, a round-number test, bunching below the simplified acquisition threshold and recipient concentration.

1. Results on agency totals

Test	Result	Reading
First-digit test, all three amounts pooled (n = 301)	Chi-square p = 0.024; MAD 0.0195 ("nonconformity" on Nigrini's scale); spike at digit 3	**A false positive, and a useful one.** The three amounts per agency are near-copies of each other, so the 301 values are really about 100 independent numbers counted three times; that inflates the chi-square. Benford tests assume many independent transactions
First-digit test, outlays only (n = 101)	Chi-square p = 0.20; MAD 0.031	No evidence against Benford. The MAD still reads "nonconformity" because Nigrini's MAD ranges assume thousands of records; at n = 101, MAD is mostly noise. This is why the handbook sets a minimum of 1,000 records
Concentration	Gini 0.94; top five agencies 85% of outlays, top ten 94%	Expected: Social Security, health, Treasury and defense dominate. Context for the scorecard, not a fraud signal
Reconciliation	No agency with more than $100M in budget authority shows obligations or outlays more than 5% above it	Clean at this level. Agency-level totals are reconciled before publication; the test earns its keep at the program and award level

The page shows the observed and expected digit distributions side by side so a viewer can see what a "pass" looks like before seeing a "flag".

2. Award-level demo (when run with network access)

`fetch_usaspending.py` pulls the most recent contract awards for one agency and year (sorted by date, not size, so the sample is not only the largest awards). `run_demo.py` then reports, for each file:

- first-digit test (when there are at least 100 awards);
- share of awards that are round thousands;
- bunching below the simplified acquisition threshold (when at least 200 awards fall between half and one and a half times it);
- number of recipients, Herfindahl index and the top-ten recipient share.

No recipient is named on the page. Award-level results are statistics on public records, for showing the method, not leads.

3. How to refresh

```
python3 analytics/roi/model.py
python3 analytics/demo/run_demo.py
```




```
# with network access to api.usaspending.gov:
python3 analytics/demo/fetch_usaspending.py \
  --agency "Department of Veterans Affairs" --fy 2025
python3 analytics/demo/run_demo.py
```

Or run the **Demo data** workflow from the repository's Actions tab; it uploads `docs/demo/data/` as an artifact to commit.

4. Next demos

1. CMS hospice enrollment and owner files through the network scorer (HHS-1), once the public files are pulled.
2. SAM.gov exclusions matched to USAspending recipients (cross-agency matching, business side).
3. Single Audit repeat findings for the largest pass-through programs.

